

SHUTAO ZHENG

+1 (416)801-5771 ♦ <https://zhengshutao.github.io> ♦ stzheng@uwaterloo.ca

EDUCATION

University of Waterloo

Bachelor of Mathematics

Computational Mathematics Major & Combinatorics and Optimization Minor & Computing Minor

Relevant Courses: Algorithm Problem Solving and Data Abstraction, Data Types and Structures, Object-Oriented Software Development, Data-Intensive Distributed Computing, Applied Cryptography Game Theory

Waterloo, Ontario

September 2018 - August 2023

WORK EXPERIENCE

Web Developer | AEC Daily Inc | Newmarket, ON | Co-op Jan. - Apr. 2023 Fulltime May 2024 - Now

- Designed and implemented an object-oriented reporting framework with extensible API design and multi-format output (Excel/HTML), significantly improving system maintainability, reducing code duplication, and enhancing enterprise reporting efficiency.
- Discovered, reported, and resolved a critical security vulnerability in the admin authentication system, where access was granted solely based on email domain without verifying ownership. Led risk analysis and implemented stricter validation protocols, significantly improving system security.
- Extended and maintained a gettext-based internationalization system across Smarty/PHP templates to support market expansion into Europe, managing translation extraction and locale-specific rendering for French and Spanish.
- Developed a Recommended Courses feature that dynamically suggested relevant content by querying the course database using PostgreSQL and integrating with front-end templates via Smarty and PHP, enhancing user engagement and personalized learning paths.

Software Engineer | Lifion by ADP | New York, NY | May - Aug. 2022

- Redesignated and incorporated new features into an internal flowchart application with Node.js, resulting in improved output information and a better user experience.
- Developed new logic operators for the internal flowchart application, increasing its functionality for users.
- Implemented integration testing via Jenkins, ensuring coverage of over 90% of newly introduced features and guaranteeing the robustness of the application.

Quality Assurance Engineer | NetDragon Websoft Holdings | Fuzhou, China | Sept. 2020 - Aug. 2021

- Developed an automated testing system for ActivPanel by Promethean, an educational interactive display deployed in over 1 million classrooms worldwide.
- Played a key role in three product launches by creating an optimized test workflow in JIRA, facilitating efficient test case management, risk assessment automation, and generation of insightful test reports.

PROJECTS

CELPip Exam Practice Platform www.deercelpip.com

JavaScript • Python • PostgreSQL • Git

- Built a production-ready English exam practice platform with 30+ RESTful APIs (Flask), delivering test content, handling user submissions, and integrating AI models for automated scoring, personalized feedback, and speech-to-text transcription.
- Designed optimized PostgreSQL schemas with indexing, transaction management, and backup strategies, supporting thousands of students with sub-second query response times.
- Enhanced security with JWT authentication, PBKDF2 password hashing, Fernet encryption, SQL injection/XSS prevention, and GDPR-compliant data handling.
- Optimized performance through CDN integration, caching, and compression, reducing load time by 60% and scaling reliably to 500+ concurrent users.

Aya Calendar (Couples Calendar App)

Flutter • Dart • Swift/WidgetKit • CloudBase Serverless

- Designed and implemented a local-first offline architecture using Drift (SQLite) with incremental sync to Tencent CloudBase, reducing cloud read requests by [X]% while maintaining full offline availability and conflict resolution.
- Built and maintained 3 independent serverless functions on Tencent CloudBase, handling authentication, data sync, and business logic for a multi-module couples productivity app.
- Extended platform reach with native iOS Home Screen Widgets (Swift/WidgetKit) and automated the release pipeline via Xcode Cloud CI, delivering a fully cross-platform, production-ready product end-to-end.

SKILLS

Programming Language: Python, C, C++, PHP, JavaScript/Node.js, HTML/CSS, Spark, R

Development Tools: Linux, VS Code, Docker, PostgreSQL, Jira, Jenkins, Bugzilla